

n8n Multi-Agent System

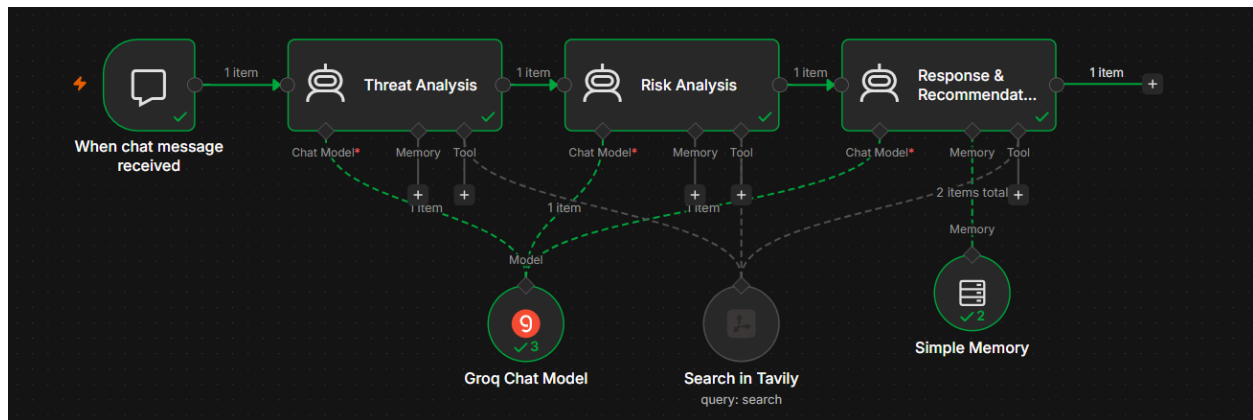
Implementation Documentation

1. n8n Workflow Setup

A new n8n workflow was created for implementing a sequential multi-agent cybersecurity system. The workflow uses three specialized AI agents that process the user's cybersecurity query sequentially.

Workflow sequence:

Chat Trigger → Threat Analysis Agent → Risk Analysis Agent → Response & Recommendations Agent → Final Response



2. Chat Trigger Configuration

This was configured as the entry point of the workflow. It receives the user's cybersecurity query and passes it to the first agent. The workflow was also published so that the chatbot could be accessed through the production chat interface.

When chat message received

Parameters Settings Open chat

Chat URL

`http://localhost:5678/webhook/7edb5c1b-58f0-4b64-9425-9753c0ca036a/chat`

☒ Make Chat Publicly Available

Mode

Hosted Chat

Chat will be live at the URL above once this workflow is published. Live executions will show up in the 'executions' tab

Authentication

None

Initial Message(s)

Hi there! 🙌
My name is Cyder. How can I assist you today?

Options +

Input Placeholder

How can I help you today?

3. Threat Analysis Agent

The first AI agent is responsible for analysing the cybersecurity threat.

System Instructions

"You are a Threat Analysis Agent in a sequential cybersecurity multi-agent system. Analyze the cybersecurity incident or problem provided in the input.

Your task is to:

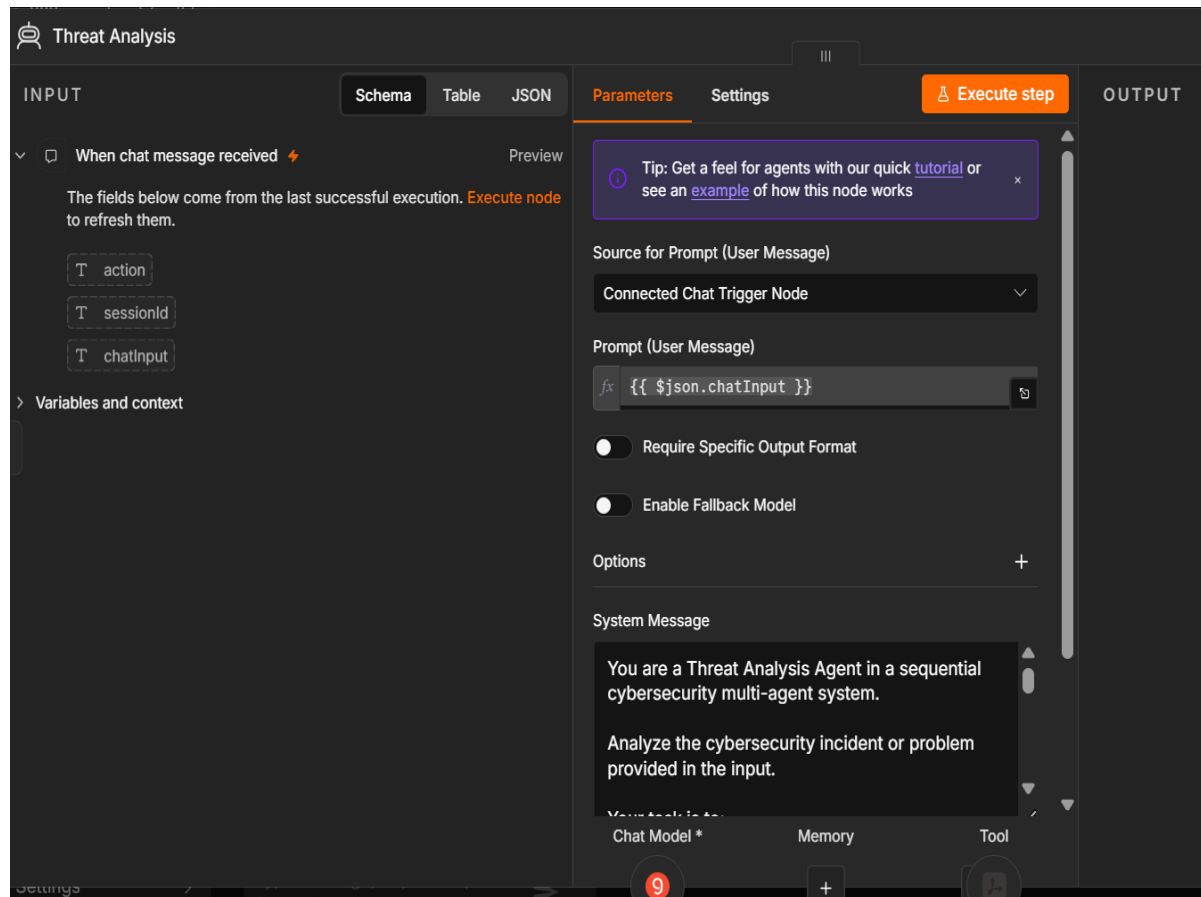
1. Understand the incident or security problem.
2. Identify the relevant threat(s).
3. Identify possible attack vectors or exploitation methods.
4. Identify vulnerabilities or weaknesses involved.
5. Explain the likely security consequences.

You have access to the Tavily Search tool.

Use Tavily Search when current, external, or verifiable cybersecurity information is required.

Do not perform unnecessary searches when the available information is sufficient. Provide a structured and technically accurate threat analysis. Your output will be passed directly to the next Risk Analysis Agent. Focus only on threat analysis and do not provide the final response to the user."

The output of this agent is passed to the Risk Analysis Agent.



4. Risk Analysis Agent

The second agent receives the threat analysis produced by Agent 1. Its purpose is to determine the potential impact and security risks associated with the identified threats.

System Instructions

"You are a Risk Analysis Agent in a sequential cybersecurity multi-agent system. You receive the threat analysis produced by the previous Threat Analysis Agent.

Your task is to:

1. Review the identified threats, vulnerabilities, and attack vectors.
2. Determine the potential security impact.
3. Identify business and operational risks.
4. Assess the possible consequences for confidentiality, integrity, and availability.
5. Identify the key risks that should be prioritized.
6. Explain the reasoning behind the identified risks.

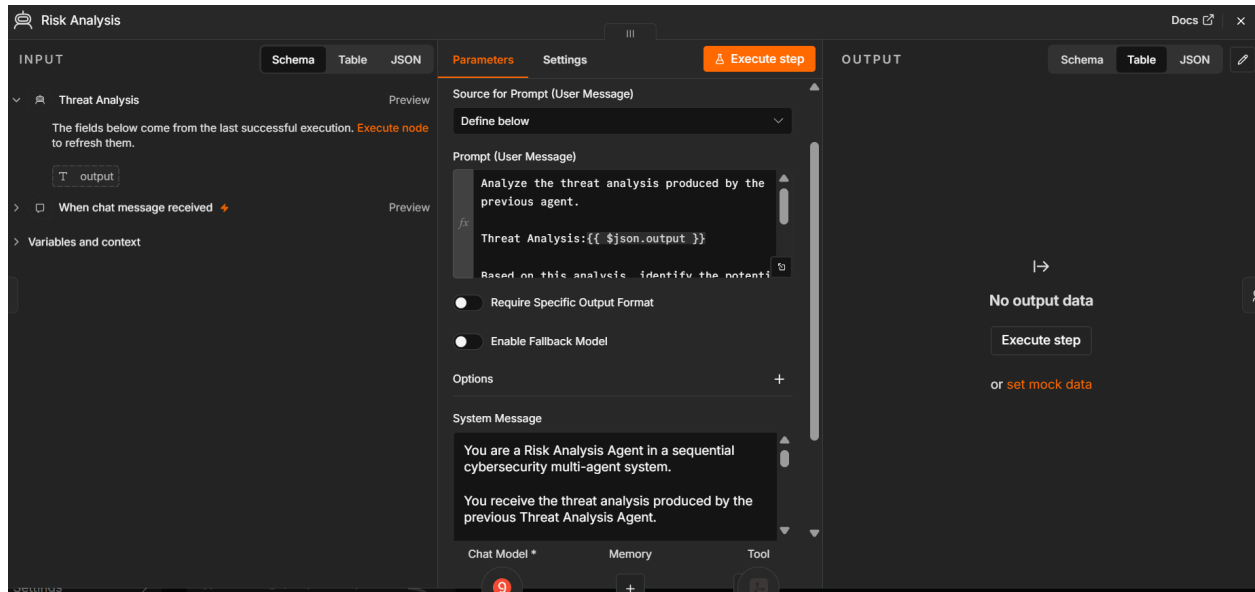
You have access to the Tavily Search tool.

Use Tavily Search only when current, external, or verifiable cybersecurity information is required. Do not perform unnecessary searches when the available information is sufficient.

Provide a structured and technically accurate risk analysis. Your output will be passed directly to the Response & Recommendations Agent. Do not provide the final response to the user.”

Dynamic Input

The agent receives the previous agent's output using: `{{ $json.output }}`



5. Response & Recommendations Agent

The third agent acts as the final response stage. It receives the risk analysis from Agent 2 and converts it into practical recommendations.

System Instructions

You are a Response and Recommendations Agent in a sequential cybersecurity multi-agent system. You receive the risk analysis produced by the previous Risk Analysis Agent.

Your task is to:

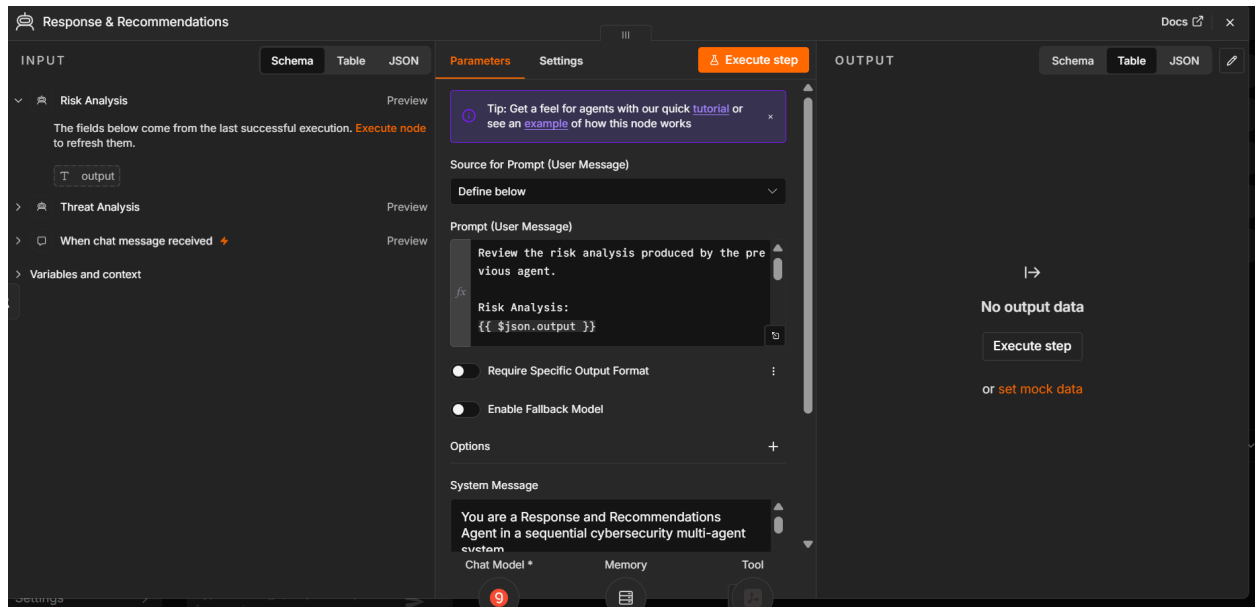
1. Review the identified risks and their potential impact.
2. Recommend practical mitigation actions.
3. Recommend relevant cybersecurity controls.
4. Suggest immediate incident response actions where applicable.
5. Suggest preventive measures to reduce the likelihood of recurrence.
6. Present the recommendations clearly, logically, and in a practical manner.

You have access to the Tavily Search tool. Use Tavily Search when current, external, or verifiable cybersecurity information is required. Do not perform unnecessary searches when the available information is sufficient. Use the threat and risk analysis provided by the previous agents as context. Produce the final response for the user.

Do not mention the internal multi-agent workflow or refer to yourself as an agent.

Dynamic Input

The Risk Analysis output is passed explicitly using: `{{ $json.output }}`

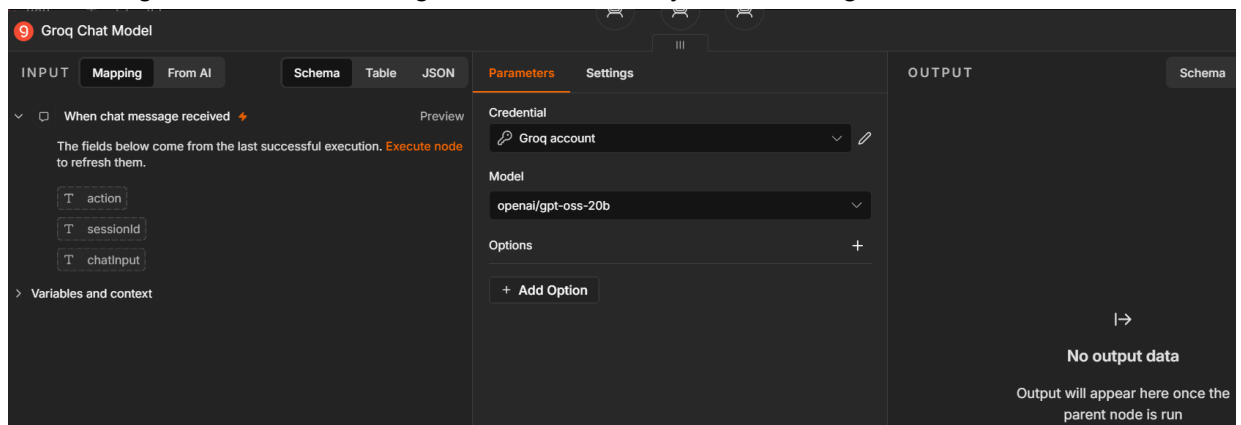


6. Groq Model Configuration

A single Groq Chat Model configuration was connected to all three agents.

Configuration Setting	Value
Provider	Groq
Model	openai/gpt-oss-20b
Usage	Shared by all three agents

Using one model configuration allows all agents to use the same underlying LLM while maintaining different roles through their individual System Messages.



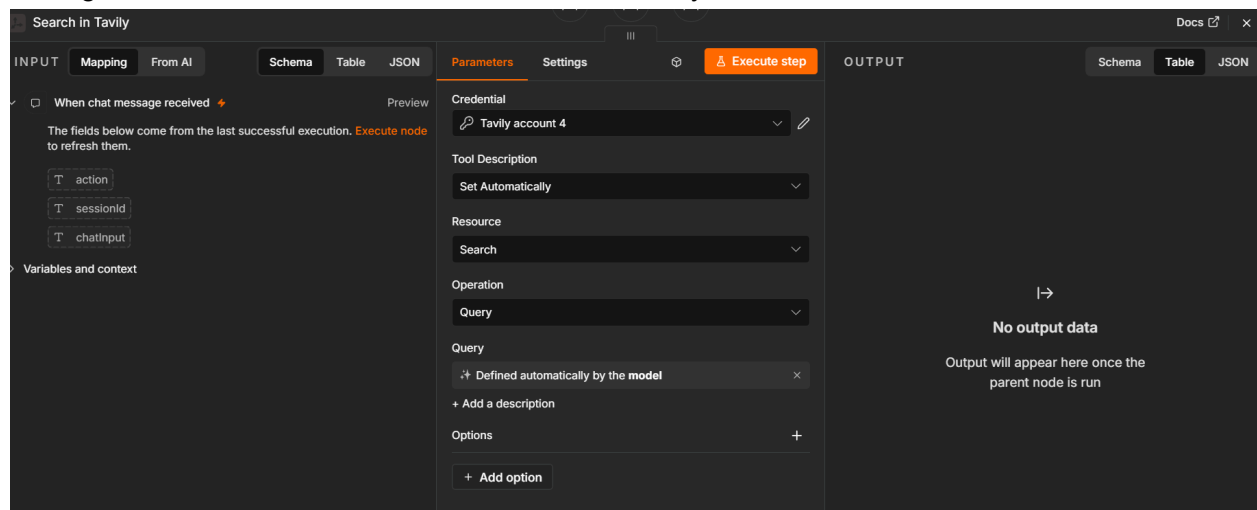
7. Tavily Search Tool

Tavily Search was added as an optional external research tool for the agents.

The agents are instructed to use Tavily when current, externally verifiable cybersecurity information is required. The tool is available to the agents without being part of the main sequential data path.

Agent → Tavily Search

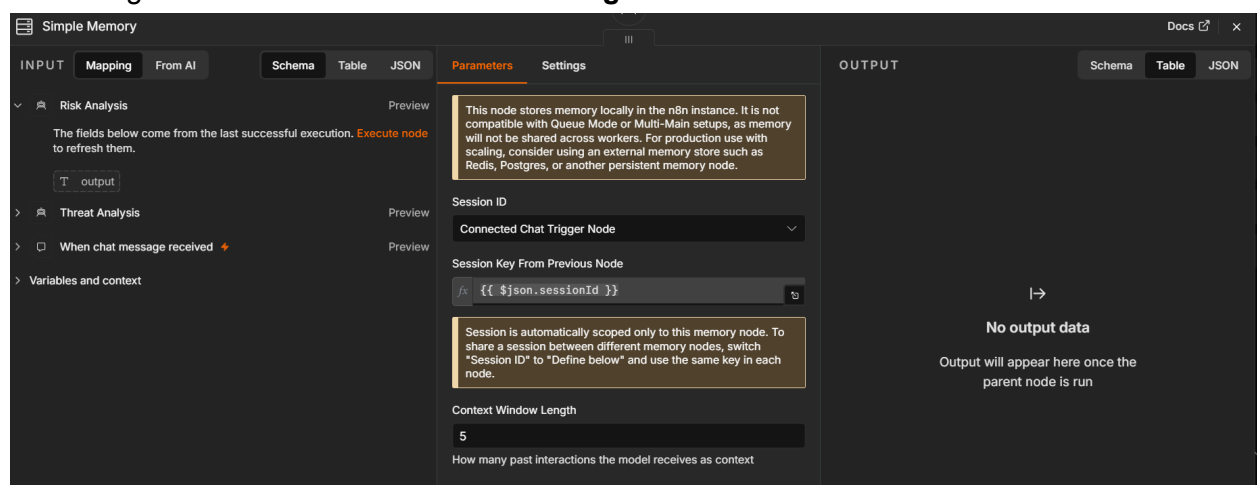
The agent decides when external search is necessary.



8. Conversational Memory

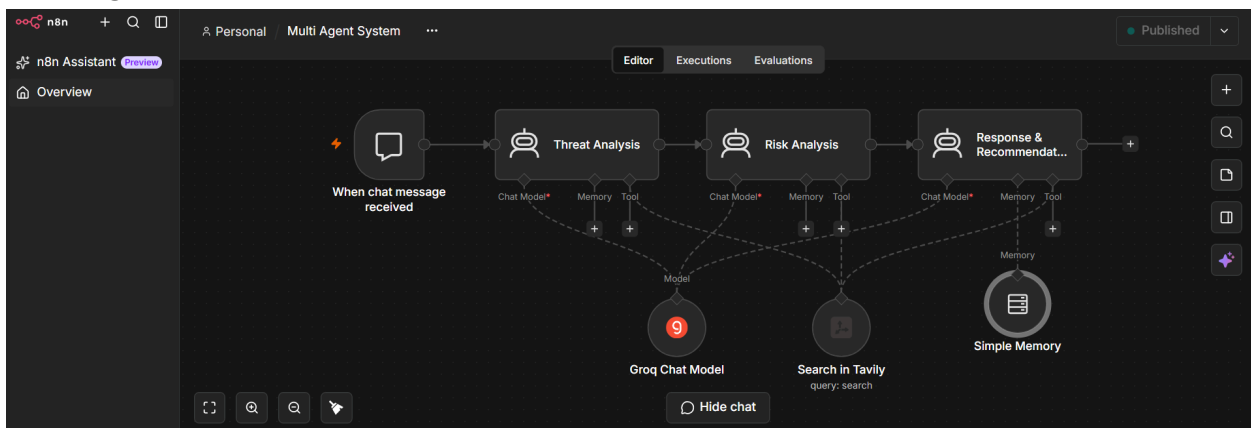
Simple Memory was added to maintain conversational context. The memory allows the chatbot to use previous messages when handling follow-up questions within the same conversation.

The configured context window was: **5 messages**



9. Sequential Multi-Agent Architecture

The final system uses three specialized agents operating sequentially. The sequential design separates the responsibilities of **threat identification**, **risk assessment**, and **response planning**.



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10. Chatbot Testing

The published chatbot was tested through the production chat interface.

Test Scenario

User Query: An employee clicked a phishing link and entered their corporate credentials. What should the organisation do?

The query was processed through the three-agent sequence, producing a final cybersecurity response.

Multi Agent System

An agent which helps with cyber security related queries

Hi there! 🙌

My name is Cyder. How can I assist you today?

An employee clicked a phishing link and entered their corporate credentials. What should the organisation do?

1. Immediate Actions (0–24 hrs)

#	Action	Target Risk(s)	Why it matters
1	Disable or lock the compromised account and enforce a password reset.	R-001, R-002, R-003	Stops the attacker's current foothold.
2	Enable MFA (adaptive, contextual) on all corporate logins (if not already in place).	R-001, R-002, R-009	Makes credential reuse ineffective and blocks most ATO attempts.

How can I help you today?